SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1108

COURSE OUTCOME COVERED

CO4: Implement object-oriented system layers using design principles and patterns, and integrate access and view layers with OODBMS *(BL3)*
Assessment Tool Weightage: 30%

QUESTIONS: 15 **Total Marks:30**

Annexure A

CLASS TEST

Code of Conduct:

I N.Nesarajan/192511386 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

CLASS TEST

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Annexure A

CLASS TEST

QUESTIONS: **Total Marks: 30**

Q1.Define architectural patterns in object-oriented systems. List any two common patterns. Explain how the Model-View-Controller (MVC) architecture improves modularity and maintainability.

[L1, L2, L3]

Q2.What is the Data Access Object (DAO) Pattern? Mention two advantages. Explain how DAO separates database-related operations from application logic.

[L2, L3, L4]

Q3.What are object-oriented design guidelines? Identify any two categories of design guidelines. Explain the role of the Liskov Substitution Principle (LSP) in developing reusable software components.

[L2, L3, L4]

Q4.Define database persistence in object-oriented applications. Mention two persistence mechanisms. Explain how an Object-Relational Mapping (ORM) framework supports persistent object management.

[L1, L2, L3]

Q5.What is an Application Layer? State two major responsibilities. Explain how the Application Layer coordinates use cases without directly implementing domain business rules.

[L2, L3, L4]

Q6.Explain the structure of a three-tier architecture consisting of Presentation, Business, and Data layers. Analyze the flow of a customer transaction through these layers in an e-commerce application.

[L2, L4, L5]

Q7.What is the role of a Business Logic Layer? List two important functions. Explain how it validates and processes requests received from the presentation layer.

[L2, L3, L4]

Q8.Explain the significance of cohesion and coupling in object-oriented design. Analyze how high cohesion and low coupling contribute to software maintainability and reusability.

[L3, L4, L5]

Q9.What is encapsulation in an object-oriented system? Explain how encapsulation protects object state and compare it with data hiding using a suitable example.

[L2, L4]

Q10.Design a layered object-oriented system for an online banking application. Identify the major layers, recommend suitable design patterns, and justify how the architecture improves security, scalability, and maintainability.

[L3, L4, L5]

Q11.What is the role of the Infrastructure Layer in object-oriented architecture? List two responsibilities. Explain how it supports technical services such as database access and external system communication.

[L1, L2, L3]

Q12.Define the Singleton Design Pattern. Mention its major characteristics. Explain its appropriate use in managing shared resources and discuss one limitation of the pattern.

[L2, L3, L4]

Q13.What is dependency injection in object-oriented design? List two benefits. Analyze how dependency injection reduces tight coupling between classes and improves testability.

[L2, L3, L4]

Q14.Explain the working of the Strategy Design Pattern. Compare the roles of Context and Strategy with a suitable real-world application example.

[L3, L4, L5]

Q15.Design a layered architecture for a Library Management System. Identify suitable layers, recommend relevant design patterns, and justify how your proposed architecture supports flexibility and maintainability.

[L3, L4, L5]

eg: E-Commerce web-site or mobile app.

(ii) Business Layer:

* Contains the application buissness logic.
* Validate orders, Calculate Price, applies discounts and processess Payment
* Acts as a bridge between the Presentation and Data layer

(iii) Data layer:

* Manages data storage and retrival.
* Communication with the database

eg: storing customer, Product and order details

~~Fork~~ Workflow:

Customer places an order → Presentation layer receives it → Business layer validates the order and calculate the Total → Data layer stores the order in database → Result is sent back to the customer

Object relational mapping supports Persistence management by maps the object to database Tables and automatically handling storing, retriving, updating and deleting object

Q5) Ans:

Application layer coordinate the application's use Cases and Controls the flow of operations.

There are two responsibilities:

* Coordinate use Cases
* Manage application workflow

It calls domain object / service to Perform bussiness operations but does not contain the Core bussiness rules itself. This keeps the system modular and maintainable.

6. Three Tier architecture:

(i) Presentation layer:

* Provide the user interface
* Accepts input from the user and display the results

Q2. Ans

Data access object is a design Pattern used to manag database operation seperately from application logic

There are two advantages

* Easier maintenance.

* Better Code reusability.

Seperation: Data access object handles operation like insert, update, Delete and retrieve data, while the application logic focuses on business task. database Code is kept seperate from business logic.

Ans

Database Persistence means storing objects / data Permanently in a database so that they can be retrived later.

There are two Persistence mechanism

* ORM (Object Relational mapping)

* JDBC

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OOAD

class Test

Q1. Model View Controller (MVC) architecture improves modularity

Ans: Architecture Patternare reusable, high level design structure that defines how components of an object oriented system are organized and interact.

Two common Patterns:

* MVC (Model View Controller)
* Layered architecture

MVC - It divides the model into:

Model - Manages the data
View - display UI
Controller - Handles user input

MVC improves the modularity and maintainability by seperating the responsibility, making the system easier to develop, Test and other Processes.